Chapter 9: Macromolecular Modeling with Rosetta

References

Papers:

- Macromolecular Modeling with Rosetta: Das and Baker, The Annual Review of Biochemistry, 2008
- ROSETTA3: An object-oriented software suite for the simulation and design of macromolecules: Leaver-Fay *et al*, Methods in Enzymology, 2011
- The Rosetta All-Atom Energy Function for Macromolecular Modeling and Design, JCTC, 2017

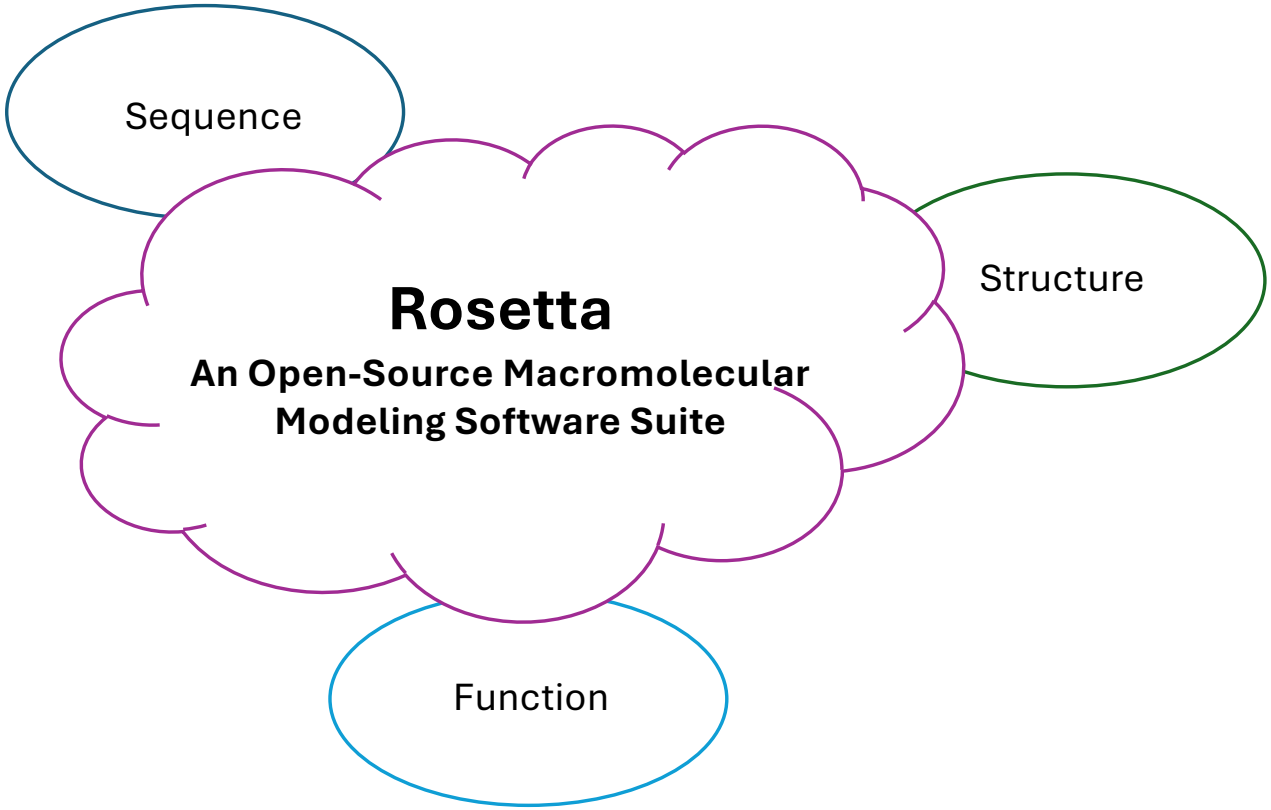
Websites:

- Foldit *https://fold.it*
- Rosetta Online Document: *https://docs.rosettacommons.org/docs/latest/Home*
- PyRosetta Online Tutorial: *https://rosettacommons.github.io/PyRosetta.notebooks/*

- What is Rosetta?
- What can Rosetta do?
- What is inside Rosetta?
- How to use Rosetta?

- What is Rosetta?
- What can Rosetta do?
- What is inside Rosetta?
- How to use Rosetta?

What is Rosetta?



What is Rosetta?

- **A comprehensive software suite for modeling macromolecular structures.**

  As a flexible, multi-purpose application, it includes tools for structure prediction, design, and remodeling of proteins and nucleic acids.

- **A unified framework for structure prediction and design**

  Rosetta boasts broadly tested scoring (energy) functions and contains an unparalleled breadth of applications from folding to docking to design.

What is Rosetta?



https://rosettacommons.org

What is Rosetta?

Rosetta is a very large software suite for macromolecular modeling.

By software suite, we mean that it is a large collection of computer code (mostly in C++, some in Python, a little in other languages), but it is not a single monolithic program.

By macromolecular modeling, we mean the process of evaluating and ranking the physical plausibility of different structures of biological macromolecules (usually protein, but nucleic acids and ligands are significantly supported and support for implicit lipid membranes is growing).

- What is Rosetta?
- What can Rosetta do?
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- How to use Rosetta?

What Can Rosetta Do?

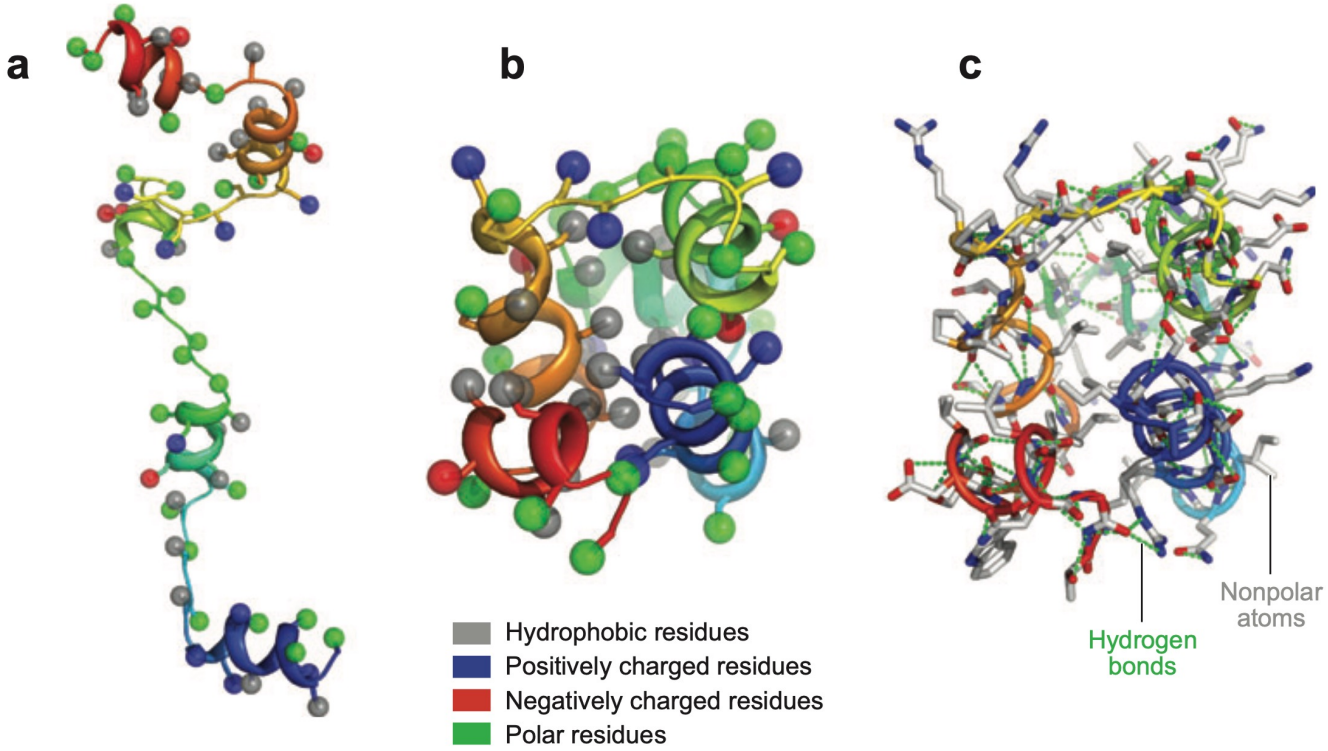
- Rosetta can build structure models.
  -- *Ab initio* structure prediction
  -- Homology modeling
  -- Structure refinement based on X-ray, NMR or EM data

- Rosetta can design proteins.
  -- Design proteins of desired topology
  -- Design protein binders
  -- Design small molecule-binding proteins
  …

- Rosetta provides a variety of tools to manipulate structure models.
  -- Perform docking simulations
  -- Redesign ligand-binding sites
  -- Redesign protein-protein interfaces
  -- Build protein chimeras
  -- Handle symmetries
  ….

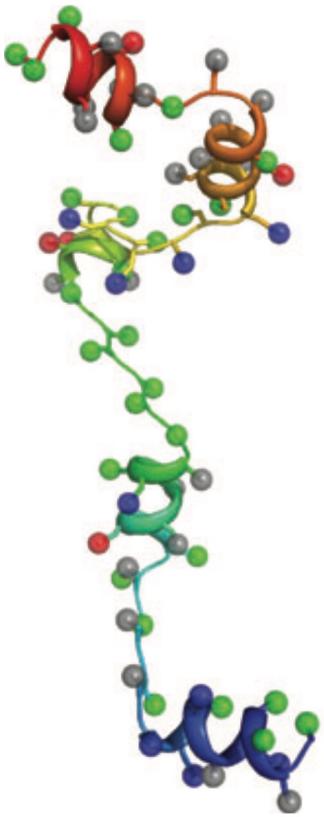
Rosetta can build structure models

- Backbone -> Sidechain
- Low Resolution -> High Resolution



Rosetta can build structure models

a



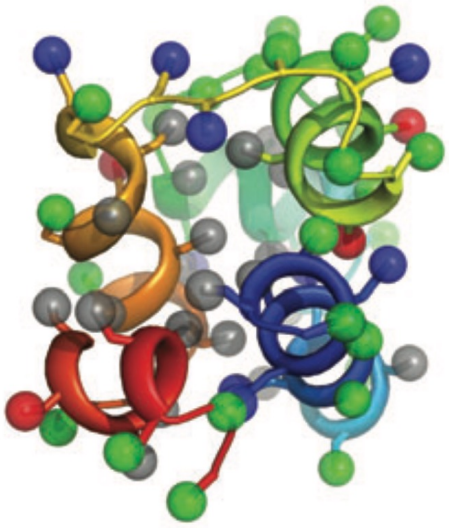
1. Build a backbone by fragment assembly

one 3-mer fragment →

```
                    phi      psi    omega
3
1kq6A T H -62.345 -36.822 177.412
1kq6A E H -67.89 -40.963 175.679
1kq6A I H -64.46 -42.656 178.586
3
1kq6A E H -67.89 -40.963 175.679
1kq6A I H -64.46 -42.656 178.586
1kq6A Y H -63.029 -41.52 178.156
3
1kq6A I H -64.46 -42.656 178.586
1kq6A Y H -63.029 -41.52 178.156
1kq6A E H -65.327 -42.397 174.125
3
1kq6A Y H -63.029 -41.52 178.156
1kq6A E H -65.327 -42.397 174.125
1kq6A F H -58.211 -46.811 177.051
3
1kq6A E H -65.327 -42.397 174.125
1kq6A F H -58.211 -46.811 177.051
1kq6A H H -53.295 -47.365 178.689
```

Rosetta can build structure models

b



Hydrophobic residues
Positively charged residues
Negatively charged residues
Polar residues

2. Represent amino acids as low-resolution blobs

Ala

```
ATOM   N    Nbb   NH1   -0.47
ATOM   CA   CAbb  CT1   0.07
ATOM   C    CObb  C     0.51
ATOM   O    OCbb  O     -0.51
ATOM   CB   CB     CT3   -0.27
ATOM   H    HNbb  H     0.31
```

Trp

```
ATOM   N    Nbb   NH1   -0.47
ATOM   CA   CAbb  CT1   0.07
ATOM   C    CObb  C     0.51
ATOM   O    OCbb  O     -0.51
ATOM   CB   CB     CT2   -0.18
ATOM   H    HNbb  H     0.31
```

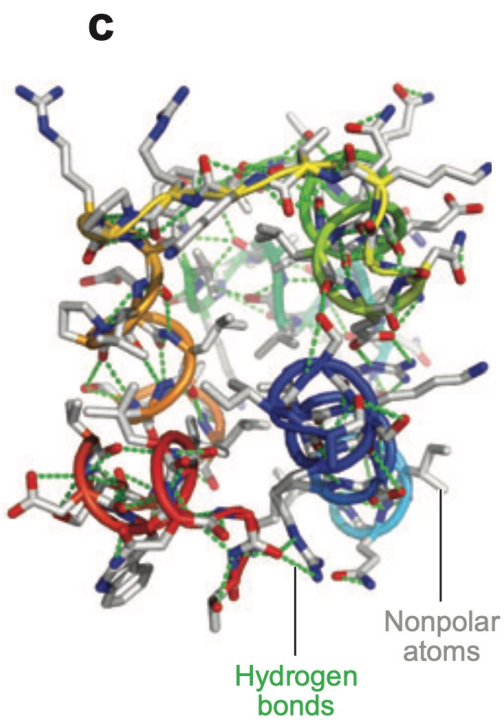
Pro

```
ATOM   N    Nbb   N     -0.29
ATOM   CA   CAbb  CP1   0.02
ATOM   C    CObb  C     0.51
ATOM   O    OCbb  O     -0.51
ATOM   CB   CB     CP2   -0.18
```

Gly

```
ATOM   N    Nbb   NH1   -0.47
ATOM   CA   CAbb  CT2   -0.02
ATOM   C    CObb  C     0.51
ATOM   O    OCbb  O     -0.51
ATOM   H    HNbb  H     0.31
```

Rosetta can build structure models



2. Represent amino acids at atomic resolution

Ala

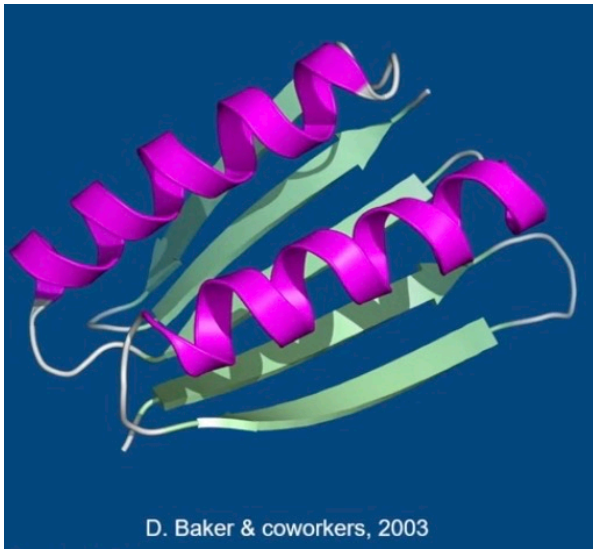
```
ATOM   N    Nbb   NH1  -0.6046255 -0.350
ATOM   CA   CAbb  CT1   0.0900506  0.100
ATOM   C    CObb  C     0.6884871  0.550
ATOM   O    OCbb  O    -0.6884871 -0.550
ATOM   CB   CH3   CT3  -0.2892500  0.000
ATOM   H    HNbb  H     0.3987955  0.250
ATOM   HA   Hapo  HB    0.1157793  0.000
ATOM  1HB   Hapo  HA    0.0964167  0.000
ATOM  2HB   Hapo  HA    0.0964167  0.000
ATOM  3HB   Hapo  HA    0.0964167  0.000
```

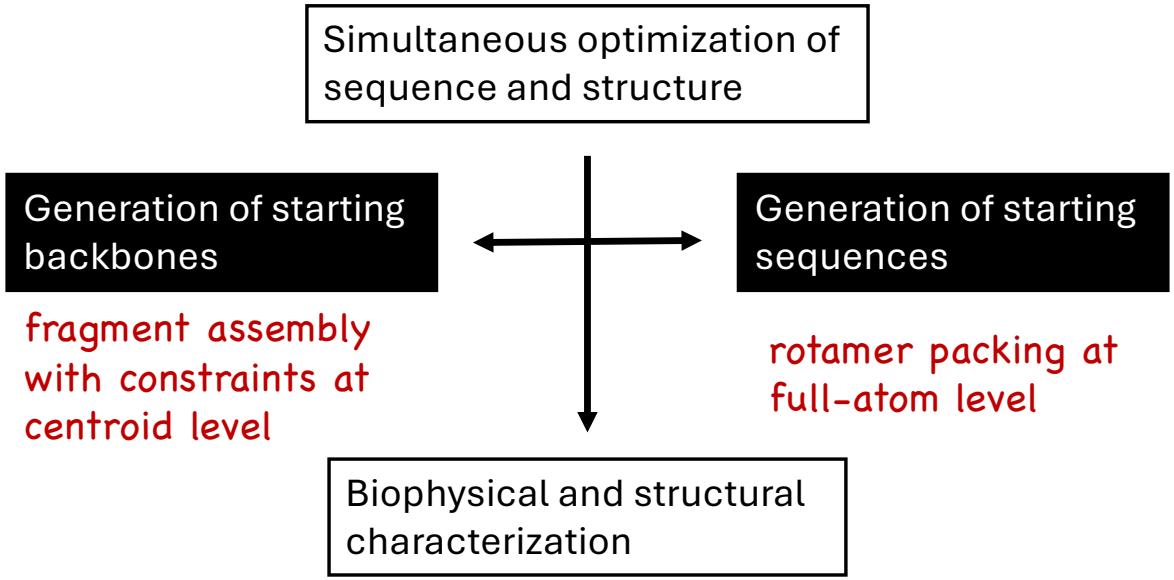
Trp

```
ATOM   N    Nbb   NH1  -0.6046255 -0.350
ATOM   CA   CAbb  CT1   0.0900506  0.100
ATOM   C    CObb  C     0.6884871  0.550
ATOM   O    OCbb  O    -0.6884871 -0.550
ATOM   CB   CH2   CT2  -0.1562226  0.125
ATOM   CG   CH0   CY   -0.0260371 -0.125
ATOM   CD1  aroC  CA    0.0303766 -0.125
ATOM   CD2  CH0   CPT  -0.0173581  0.000
ATOM   NE1  Ntrp  NY   -0.5294210 -0.400
ATOM   CE2  CH0   CPT   0.1128274  0.000
ATOM   CE3  aroC  CA   -0.0998089 -0.125
ATOM   CZ2  aroC  CA   -0.0998089 -0.125
ATOM   CZ3  aroC  CA   -0.0998089 -0.125
ATOM   CH2  aroC  CA   -0.0998089 -0.125
ATOM   H    HNbb  H     0.3987955  0.250
ATOM   HE1  Hpol  H     0.3298032  0.400
ATOM   HD1  Haro  HP    0.0998089  0.125
ATOM   HZ2  Haro  HP    0.0998089  0.125
ATOM   HH2  Haro  HP    0.0998089  0.125
ATOM   HZ3  Haro  HP    0.0998089  0.125
ATOM   HE3  Haro  HP    0.0998089  0.125
ATOM   HA   Hapo  HB    0.1157793  0.000
ATOM  1HB   Hapo  HA    0.0781113  0.000
ATOM  2HB   Hapo  HA    0.0781113  0.000
```

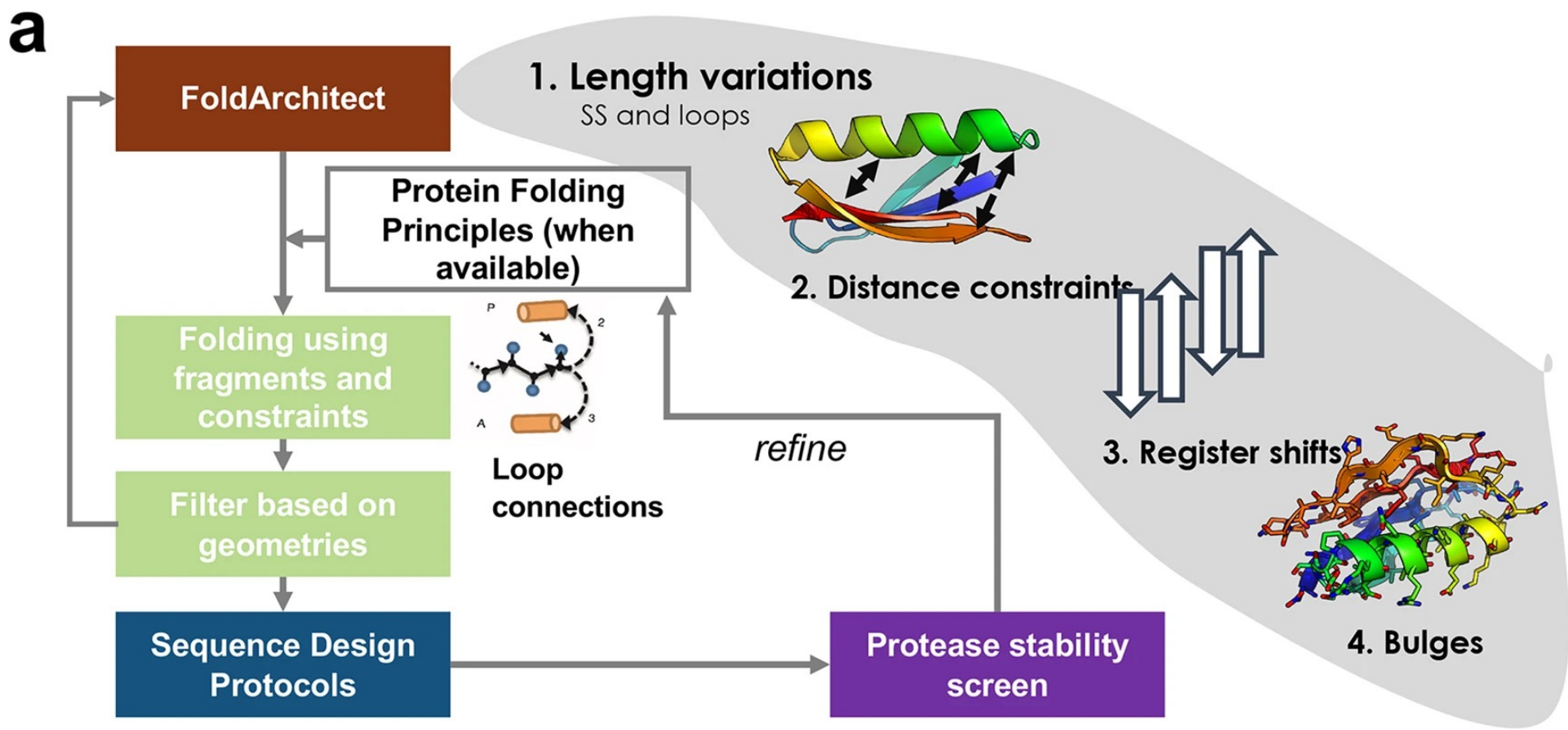
Rosetta can design proteins

Top7





Rosetta can design proteins



What Can Rosetta Do?

- **Rosetta can build structure models.**
  -- *Ab initio* structure prediction
  -- Homology modeling
  -- Structure refinement based on X-ray, NMR or EM data

  - Fragment assembly
  - Centroid representation of amino acids
  - Full-atom representation of amino acids

- **Rosetta can design proteins.**
  -- Design proteins of desired topology
  -- Design protein binders
  -- Design small molecule-binding proteins
  ...

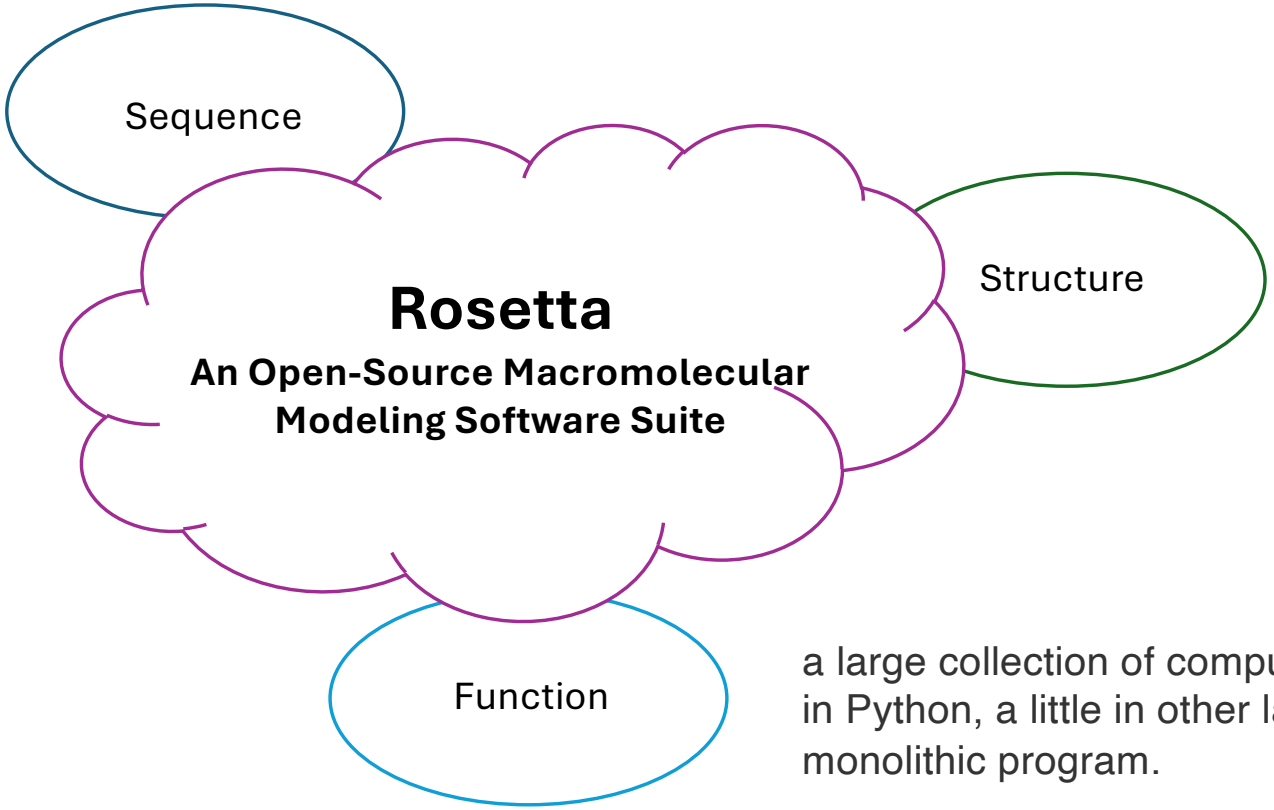
  - Fragment assembly with geometric constraints
  - Rotamer packing at full-atom level

- **Rosetta provides a variety of tools to manipulate structure models.**
  -- Perform docking simulations
  -- Redesign ligand-binding sites
  -- Redesign protein-protein interfaces
  -- Build protein chimeras
  -- Handle symmetries
  ....

Outline

- What is Rosetta?
- What can Rosetta do?
- What is inside Rosetta?
- How to use Rosetta?

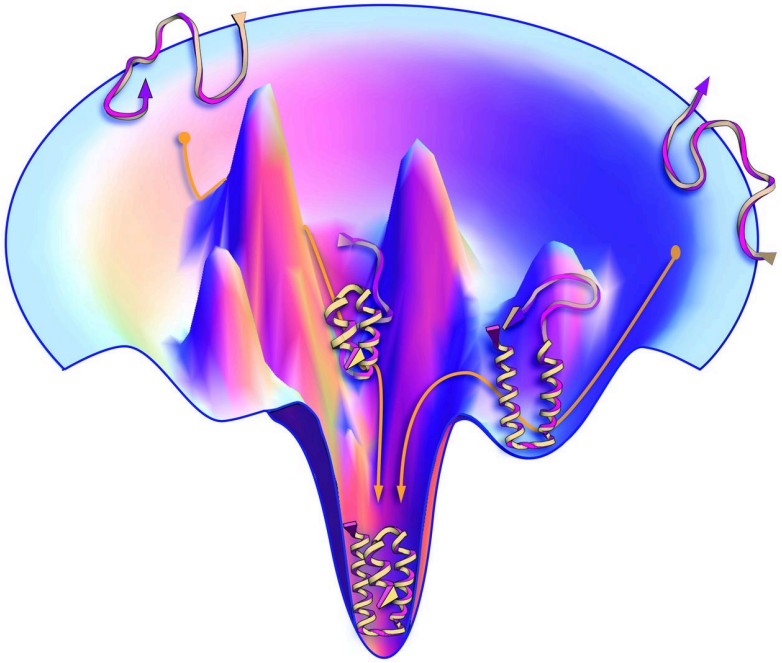
What is inside Rosetta?



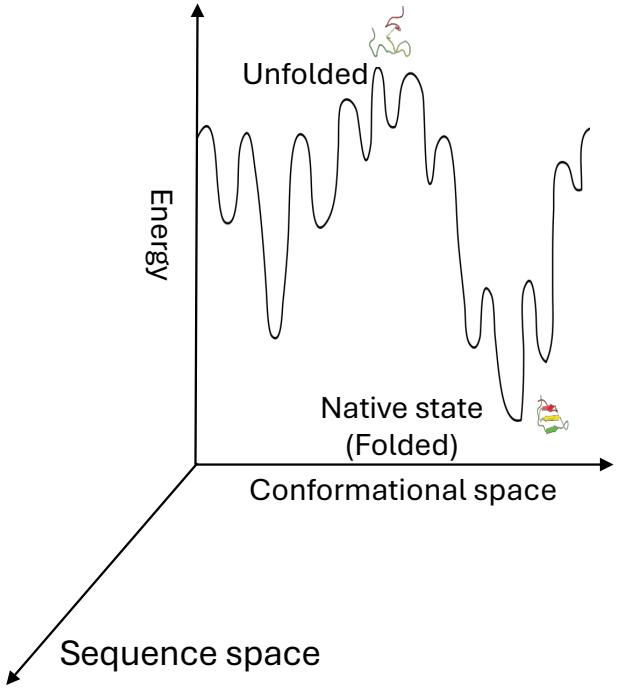
a large collection of computer code (mostly in C++, some in Python, a little in other languages), but it is not a single monolithic program.

Biophysics inside Rosetta

Protein Folding

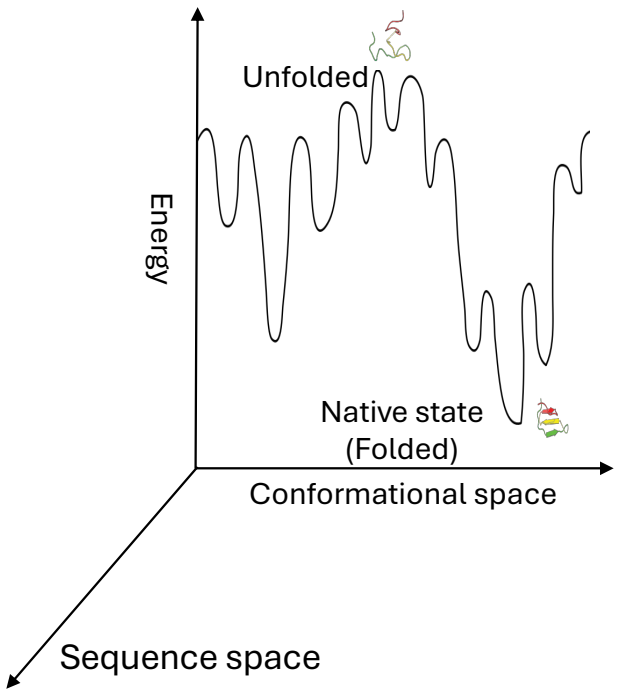


Protein Structure Prediction and Design



Biophysics inside Rosetta

Protein Structure Prediction and Design



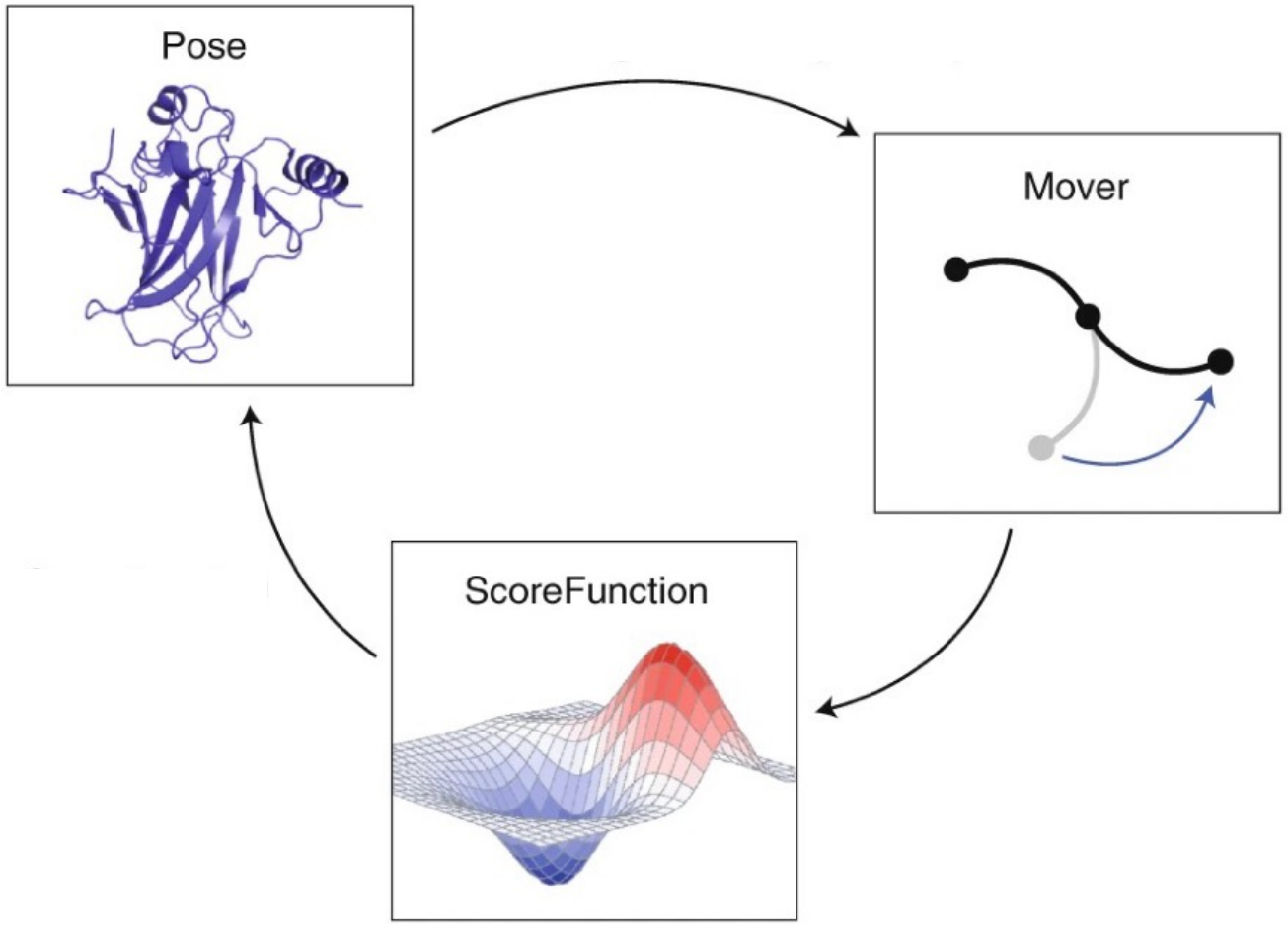
The observed conformations of folded macromolecules are almost always the lowest free-energy states.

Hence, structure prediction is generally the problem of finding the lowest-energy structure given the sequence of a protein, and design is the problem of finding the lowest-energy sequence for the target structure.

Key ingredients of molecular modeling

- Energy Function
- Conformational Sampling Strategy
- Degree of freedom

Rosetta



What is inside Rosetta?

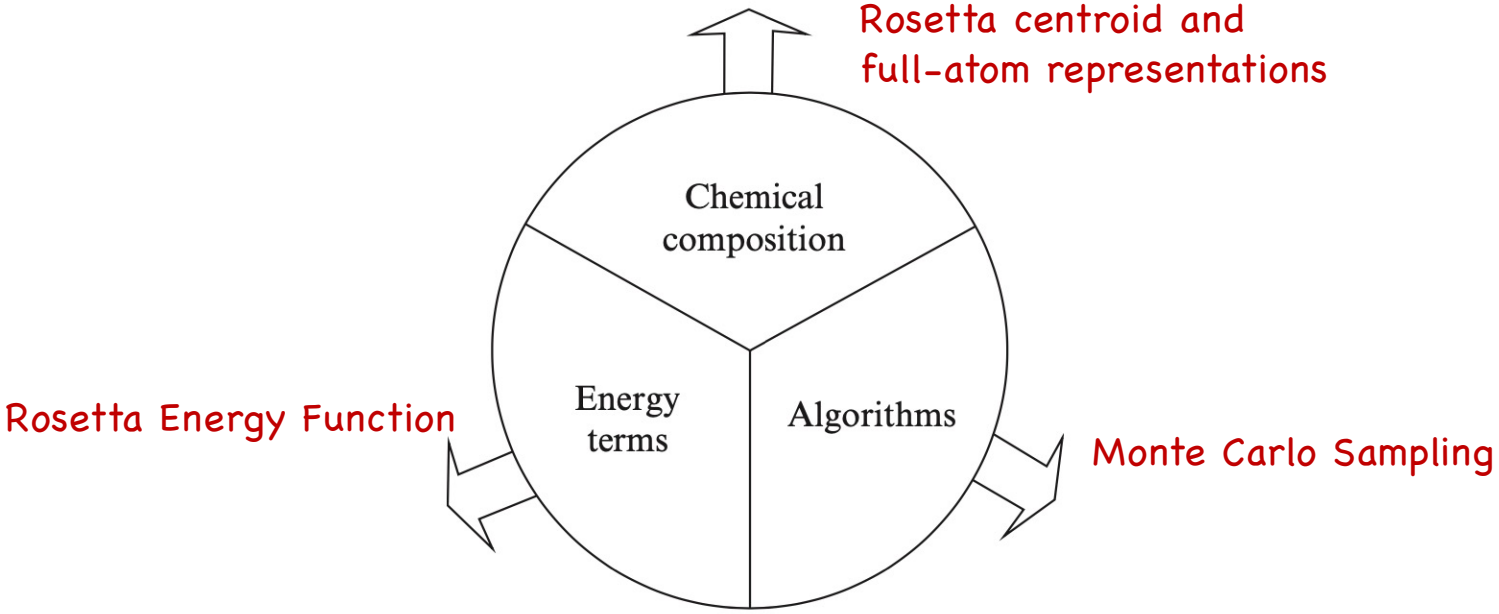
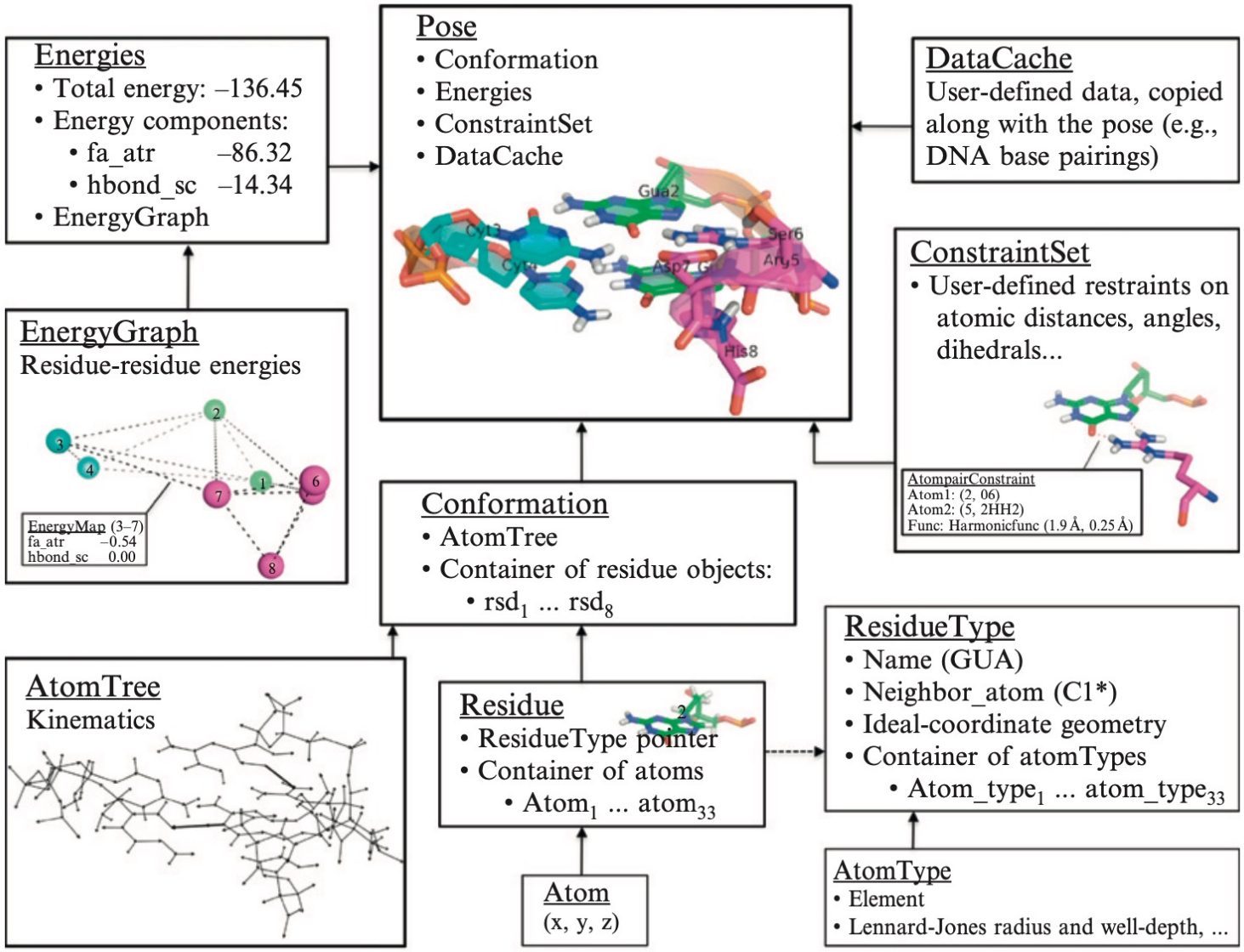
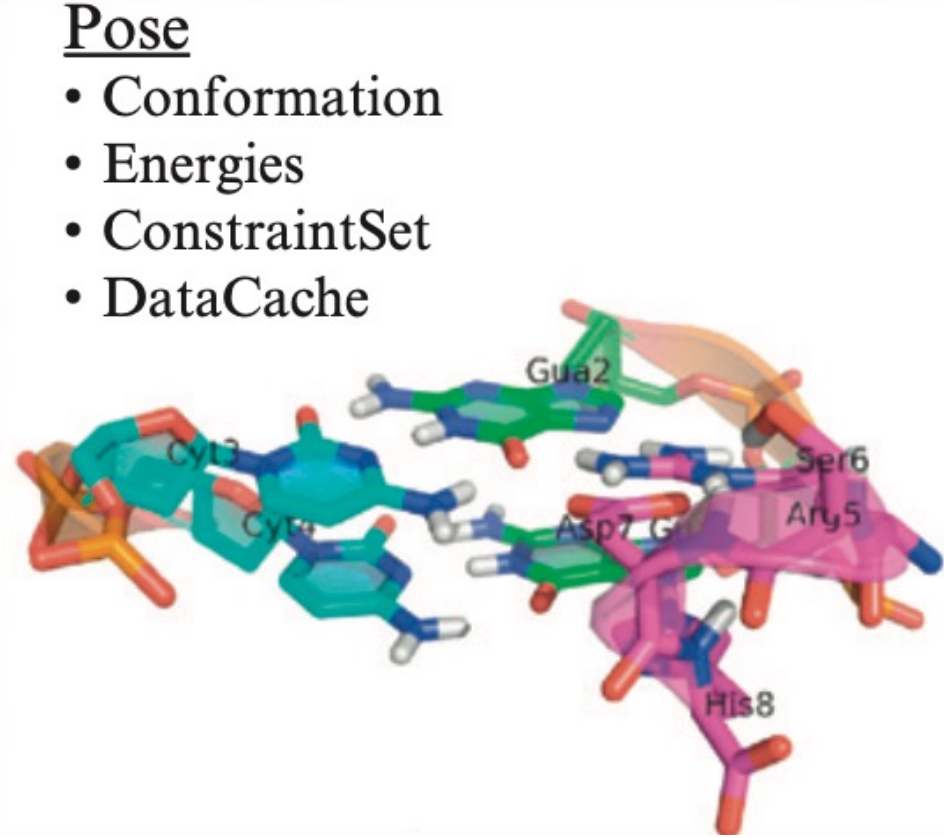


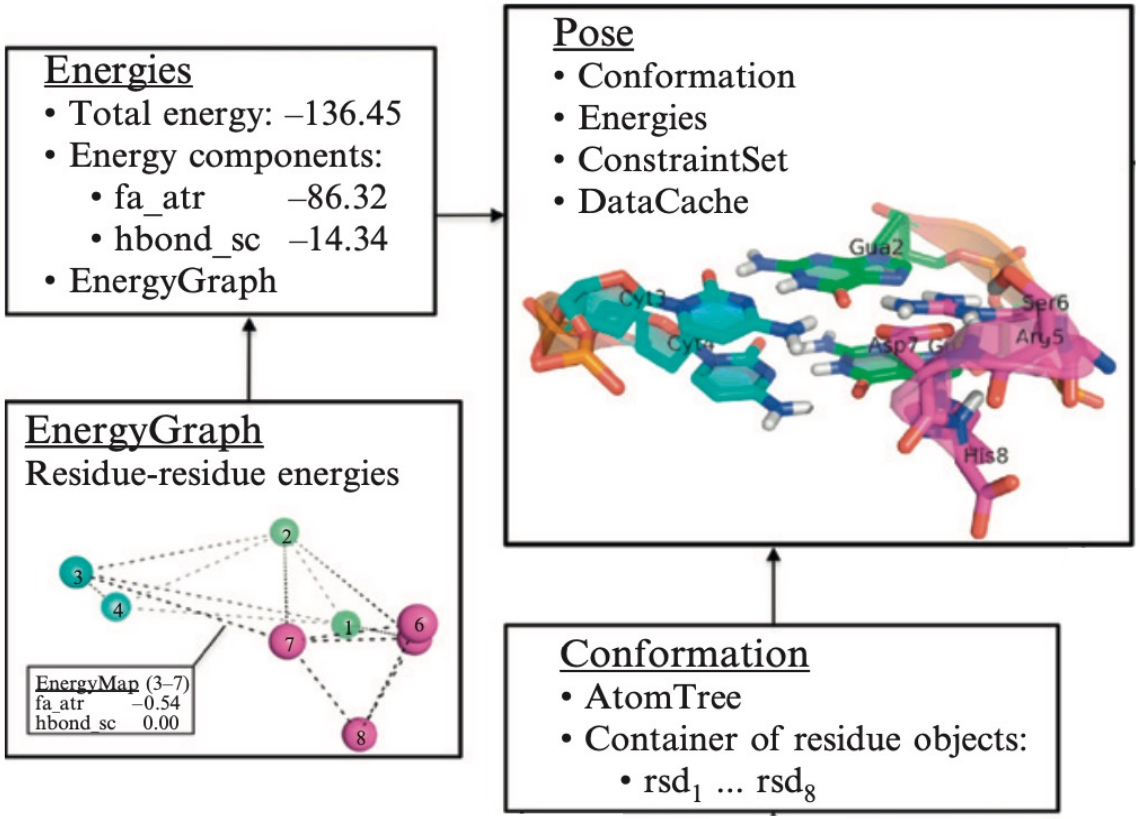
Figure 19.1 Generality Wheel. Expanding ROSETTA's functionality in one area (Energy Terms, Chemical Composition, or Algorithms) should not require an expansion to the other areas. The areas should be protected from each other through the use of generic interfaces.


Energies
• Total energy: –136.45
• Energy components:
• fa_atr –86.32
• hbond_sc –14.34
• EnergyGraph
Pose
• Conformation
• Energies
• ConstraintSet
• DataCache
DataCache
User-defined data, copied along with the pose (e.g., DNA base pairings)
EnergyGraph
Residue-residue energies
EnergyMap (3–7)
fa_atr –0.54
hbond_sc 0.00
ConstraintSet
• User-defined restraints on atomic distances, angles, dihedrals...
AtompairConstraint
Atom1: (2, 06)
Atom2: (5, 2HH2)
Func: Harmonicfunc (1.9 Å, 0.25 Å)
Conformation
• AtomTree
• Container of residue objects:
• rsd1 ... rsd8
AtomTree
Kinematics
Residue
• ResidueType pointer
• Container of atoms
• Atom1 ... atom33
ResidueType
• Name (GUA)
• Neighbor_atom (C1*)
• Ideal-coordinate geometry
• Container of atomTypes
• Atom_type1 ... atom_type33
Atom
(x, y, z)
AtomType
• Element
• Lennard-Jones radius and well-depth, ...

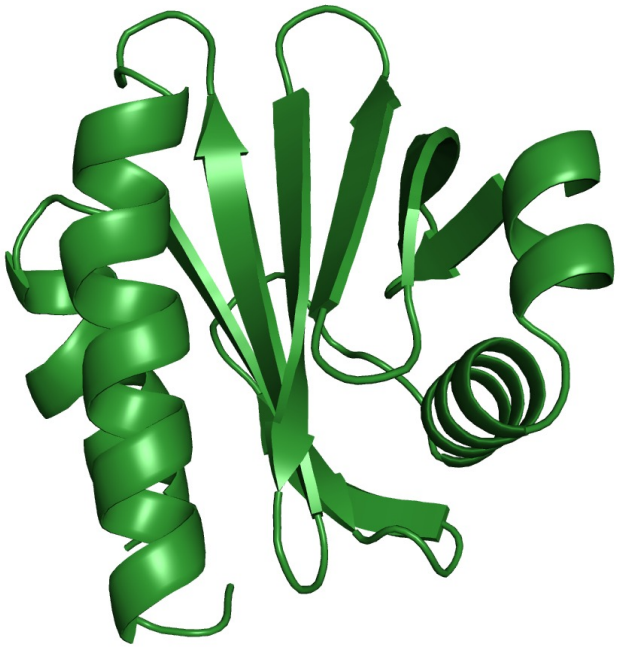
What is inside Rosetta?

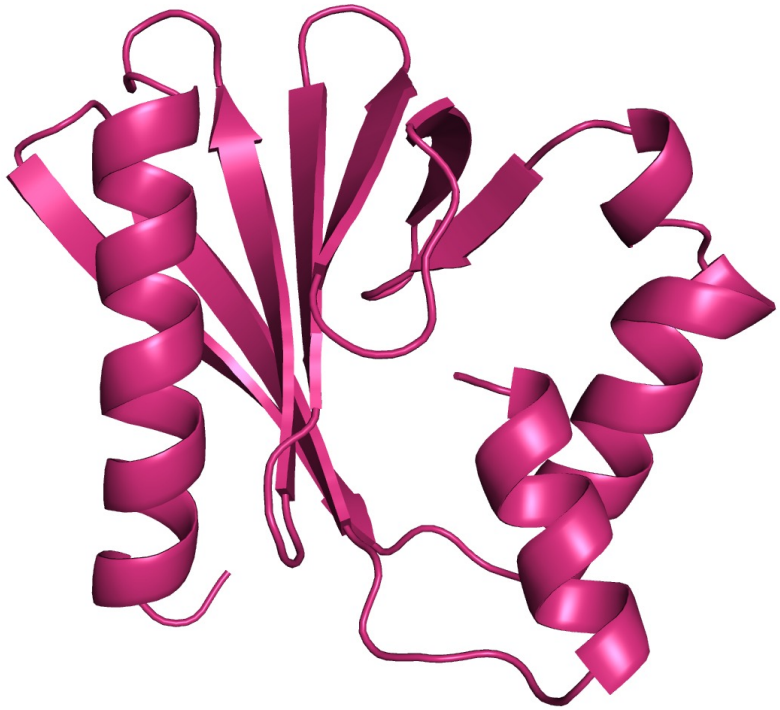


What is inside Rosetta?

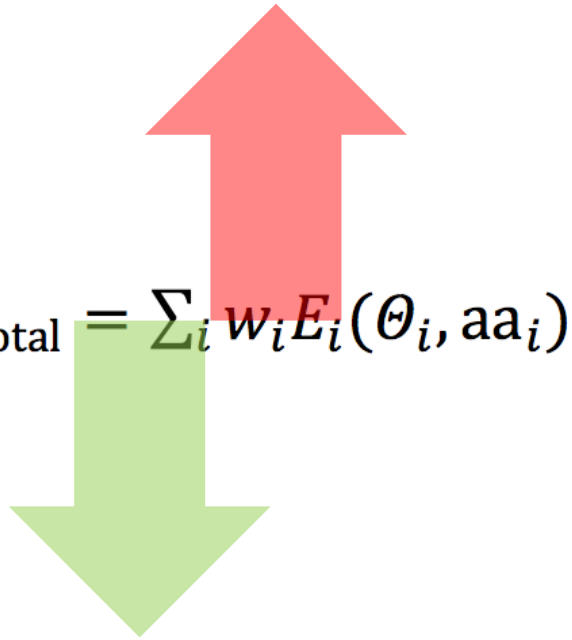


An energy function must assign a quantitative number to a model to decide whether it is “good” or “bad”





Rosetta Energy Function



$$\Delta E_{\text{total}} = \sum_i w_i E_i(\Theta_i, \text{aa}_i)$$

Energy unit: kcals/mol

Rosetta Energy Unit: REU

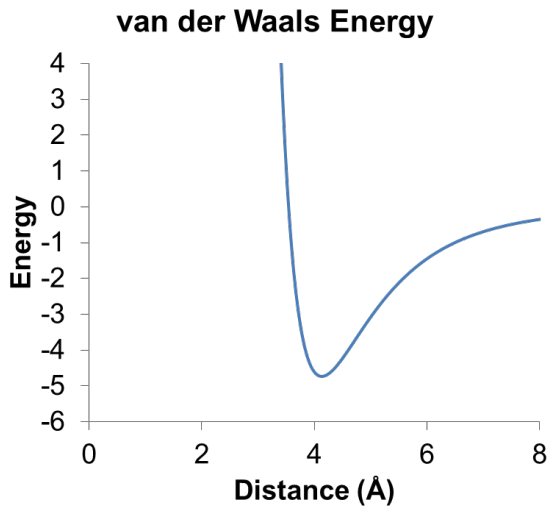
Rosetta Energy Function

- Rosetta energy function combines physics-based and knowledge-based potentials
- Rosetta energy function is a weighted linear combination of individual energy terms
- Rosetta energy function is residue-centric
- *Rosetta has two sets of energy function*

- Rosetta Combines Physics-Based and Knowledge-Based Potentials to Build the Energy Function

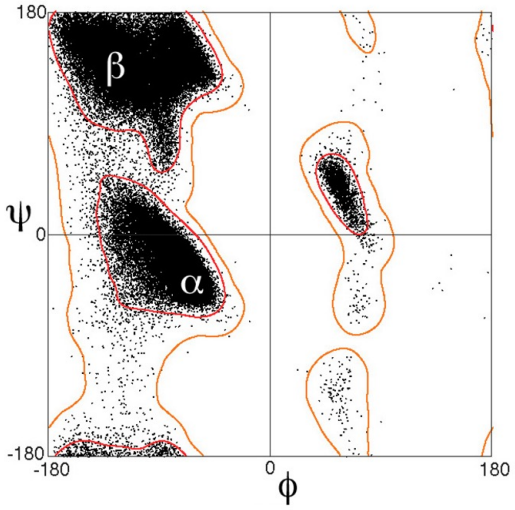
Lennard-Jones Potential

$$\sum_{i<j}\left[\frac{A_{ij}}{r_{ij}^{12}}-\frac{B_{ij}}{r_{ij}^{6}}\right]$$



Ramachandran Potential

Statistical mining of
Protein Databank (PDB)



- Rosetta Energy Function is a Weighted Linear Combination of Individual Score Terms

$$Energy = w_1 * term_1 + w_2 * term_2 + w_3 * term_3 + \ldots$$

Energy Function used for designing Top7 has 11 terms:

$$E_{protein} = W_{rot} E_{rot} + W_{aa|phi,psi} E_{aa|phi,psi} + W_{rama} E_{rama} + W_{atr} E_{atr} + W_{solv} E_{solv} + W_{pair} E_{pair} + W_{bb_hbond} E_{bb_hbond} + W_{sc_hbond} E_{sc_hbond} + W_{sc_bb_hbond} E_{sc_bb_hbond} + W_{pair} E_{pair} - E_{ref}$$

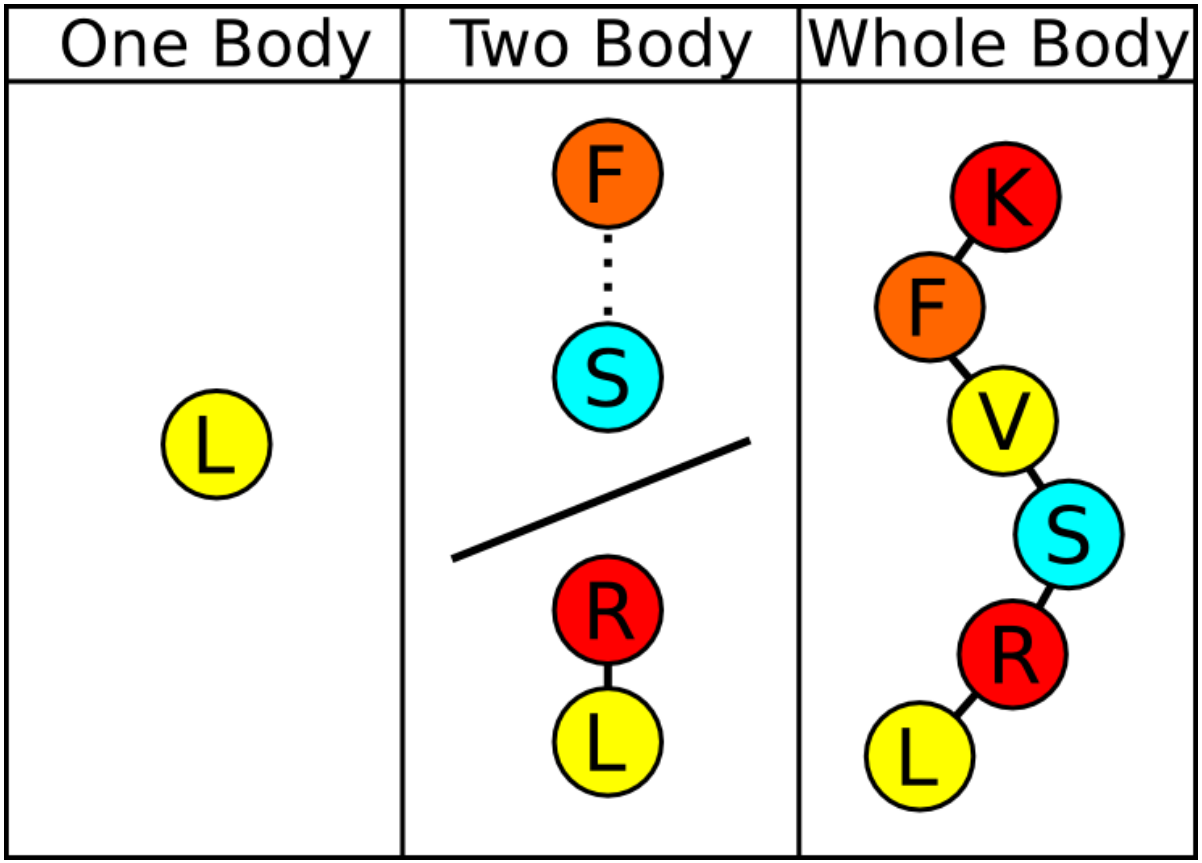
- Rosetta Energy Function is a Weighted Linear Combination of Individual Score Terms

REF2015 Energy Function

term	description	weight	units	ref(s)
fa_atr	attractive energy between two atoms on different residues separated by a distance d	1.0	kcal/mol	5, 6
fa_rep	repulsive energy between two atoms on different residues separated by a distance d	0.55	kcal/mol	5, 6
fa_intra_rep	repulsive energy between two atoms on the same residue separated by a distance d	0.005	kcal/mol	5, 6
fa_sol	Gaussian exclusion implicit solvation energy between protein atoms in different residues	1.0	kcal/mol	36
lk_ball_wtd	orientation-dependent solvation of polar atoms assuming ideal water geometry	1.0	kcal/mol	50, 71
fa_intra_sol	Gaussian exclusion implicit solvation energy between protein atoms in the same residue	1.0	kcal/mol	36
fa_elec	energy of interaction between two nonbonded charged atoms separated by a distance d	1.0	kcal/mol	50
hbond_lr_bb	energy of short-range hydrogen bonds	1.0	kcal/mol	38, 49
hbond_sr_bb	energy of long-range hydrogen bonds	1.0	kcal/mol	38, 49
hbond_bb_sc	energy of backbone−side-chain hydrogen bonds	1.0	kcal/mol	38, 49
hbond_sc	energy of side-chain−side-chain hydrogen bonds	1.0	kcal/mol	38, 49
dslf_fa13	energy of disulfide bridges	1.25	kcal/mol	49
rama_prepro	probability of backbone ϕ, ψ angles given the amino acid type	(0.45 kcal/mol)/kT	kT	50, 51
p_aa_pp	probability of amino acid identity given backbone ϕ, ψ angles	(0.4 kcal/mol)/kT	kT	51
fa_dun	probability that a chosen rotamer is native-like given backbone ϕ, ψ angles	(0.7 kcal/mol)/kT	kT	52
omega	backbone-dependent penalty for cis ω dihedrals that deviate from 0° and trans ω dihedrals that deviate from 180°	(0.6 kcal/mol)/AU	AU[a]	72
pro_close	penalty for an open proline ring and proline ω bonding energy	(1.25 kcal/mol)/AU	AU	51
yhh_planarity	sinusoidal penalty for nonplanar tyrosine χ_3 dihedral angle	(0.625 kcal/mol)/AU	AU	49
ref	reference energies for amino acid types	(1.0 kcal/mol)/AU	AU	1, 51

[a]AU = arbitrary units.

- Rosetta Energy Function is Residue-Centric



- Rosetta Energy Function is Residue-Centric

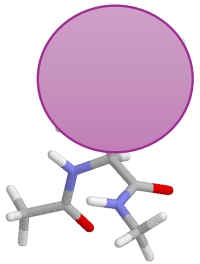
One Body	Two Body	Whole Body
Backbone - p_aa_pp - rama_prepro Side Chain - fa_dun - yhh_planarity Reference - ref	Lennard-Jones - fa_atr - fa_rep Solvation - fa_sol Hydrogen Bond - hbond_lr_bb - hbond_sr_bb - hbond_bb_sc - hbond_sc	Radius of Gyration - rg Contact Order - co Structure Alignment - hs_pair - ss_pair - sheet

* NOTE: Not all score terms are listed here

- Rosetta Energy Function has Two Modes

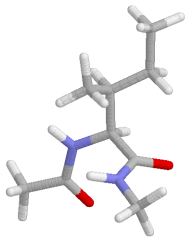
Low resolution (or centroid) mode:

Reduced atom representation (centroid)
Simple energy function
Aggressively search conformational space

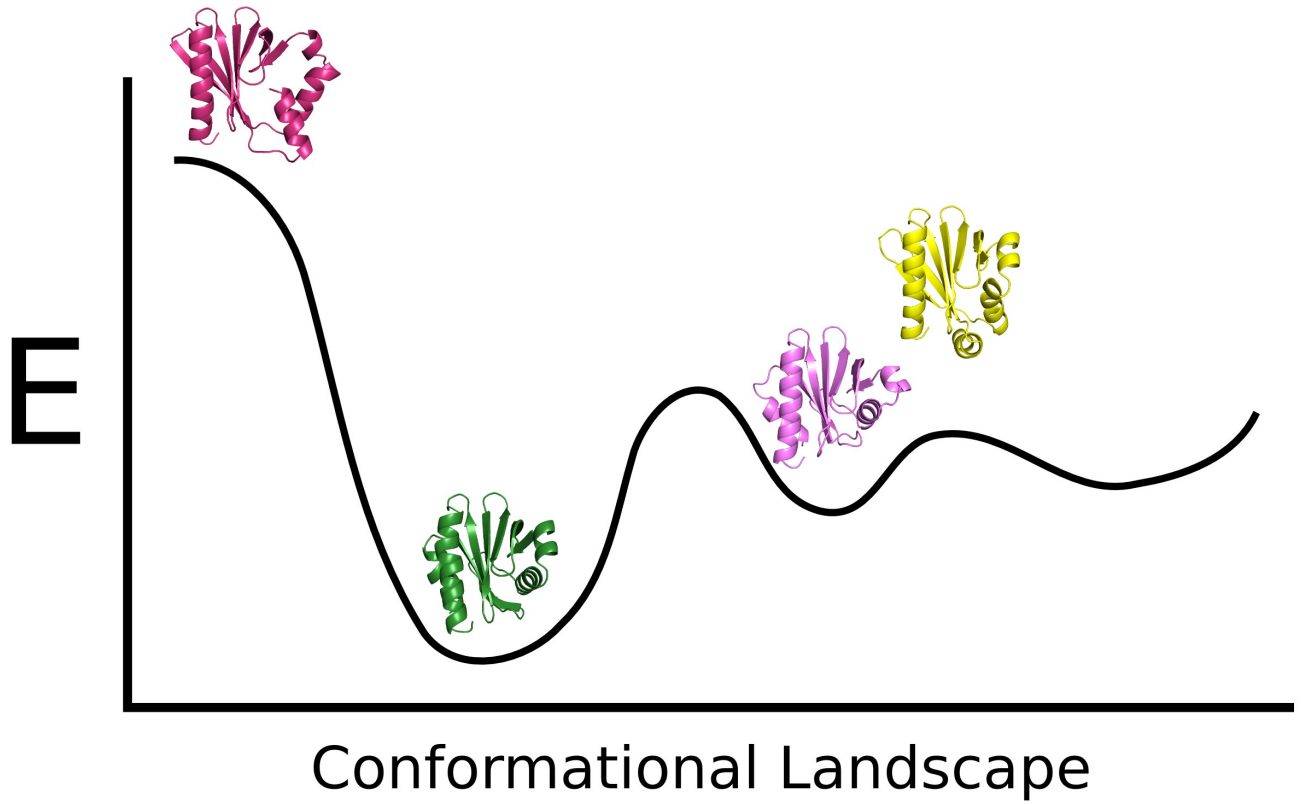


High resolution (or full atom) mode:

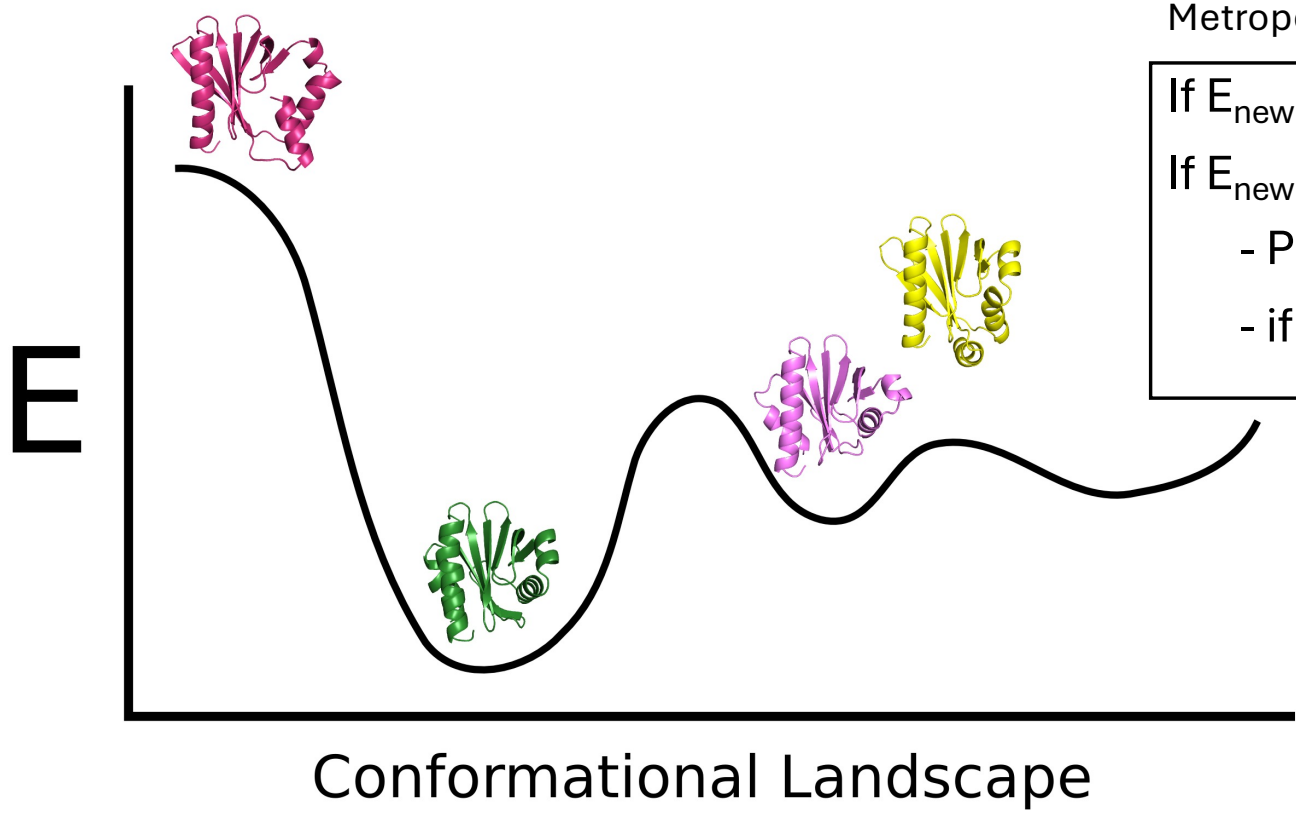
Full atom (FA)
More sophisticated energy function
“Local” search of conformational (and sequence) space



Energy Function is Central to Conformational Sampling



Metropolis-Monte Carlo Sampling Approximates Boltzmann Distribution



Metropolis Criterion

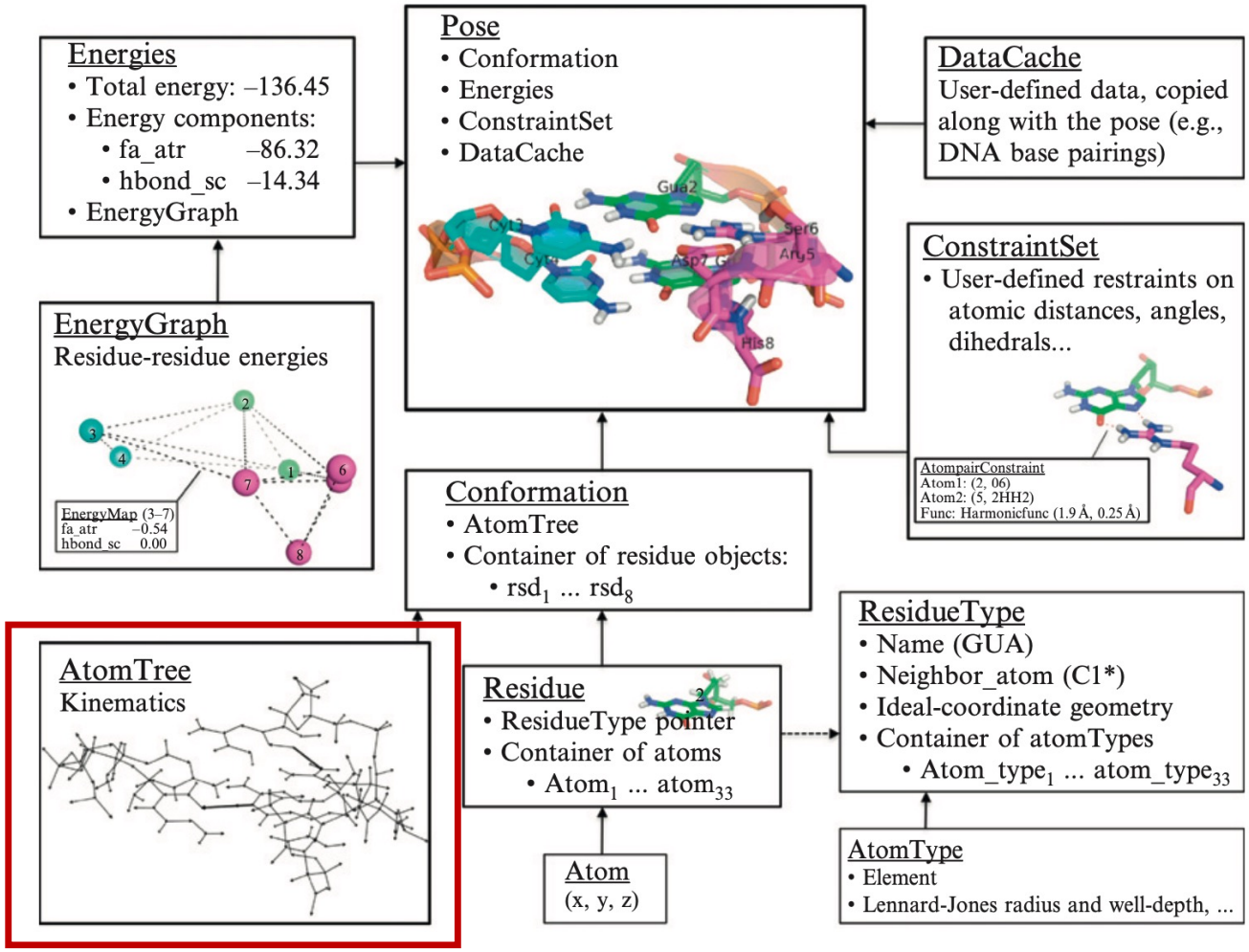
If $E_{new} < E_{old}$: Accept new structure

If $E_{new} > E_{old}$:

- Pick a random number p(0, 1)
- if $e^{[-(E_{new}-E_{old})/kBT]} > p$, accept new structure

How to move?

Kinematics in Rosetta



What is kinematics? “geometry of motion”



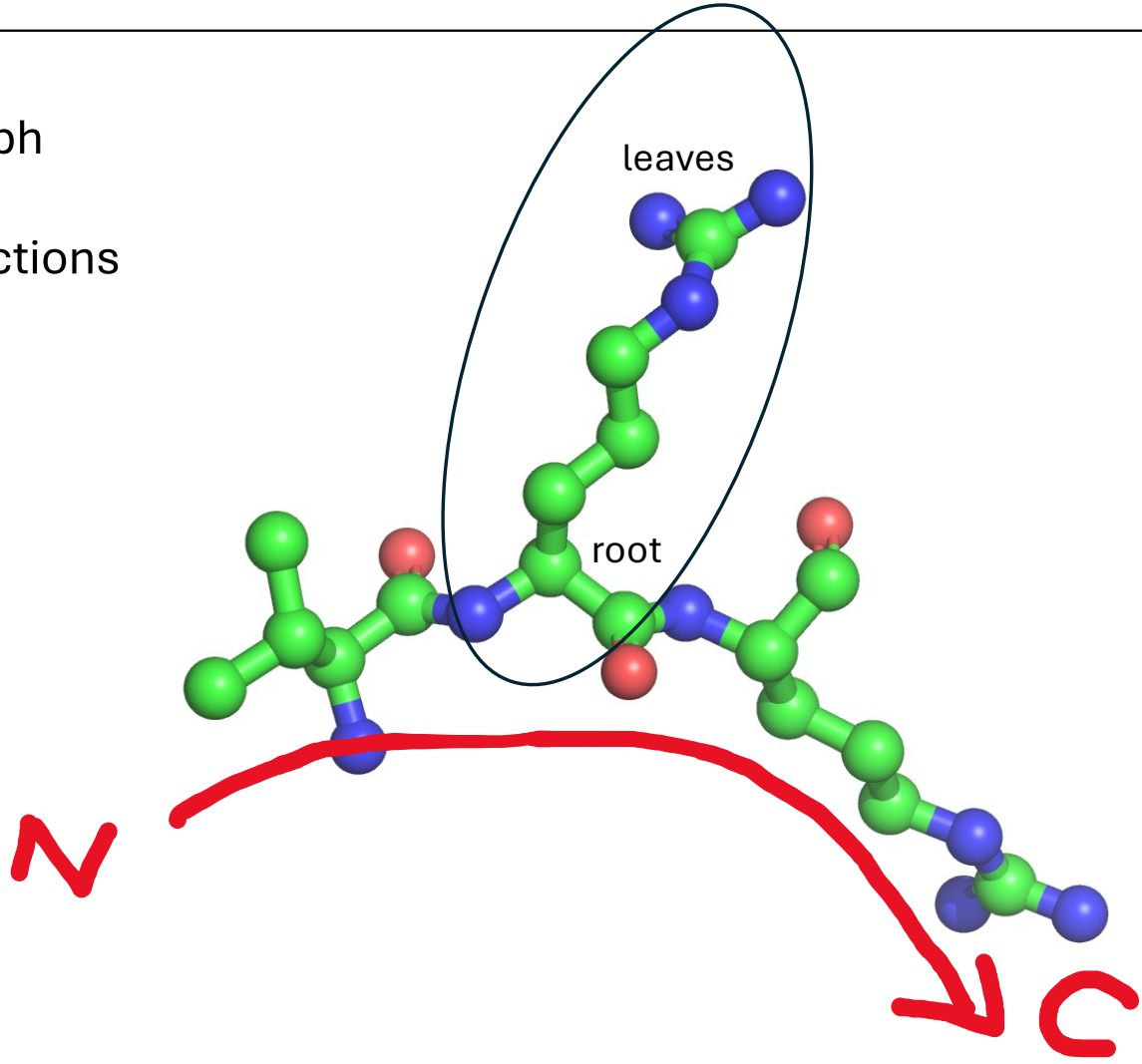
https://ocw.mit.edu/courses/8-01sc-classical-mechanics-fall-2016/pages/week-1-kinematics/week-1-introduction/

Kinematics in Rosetta

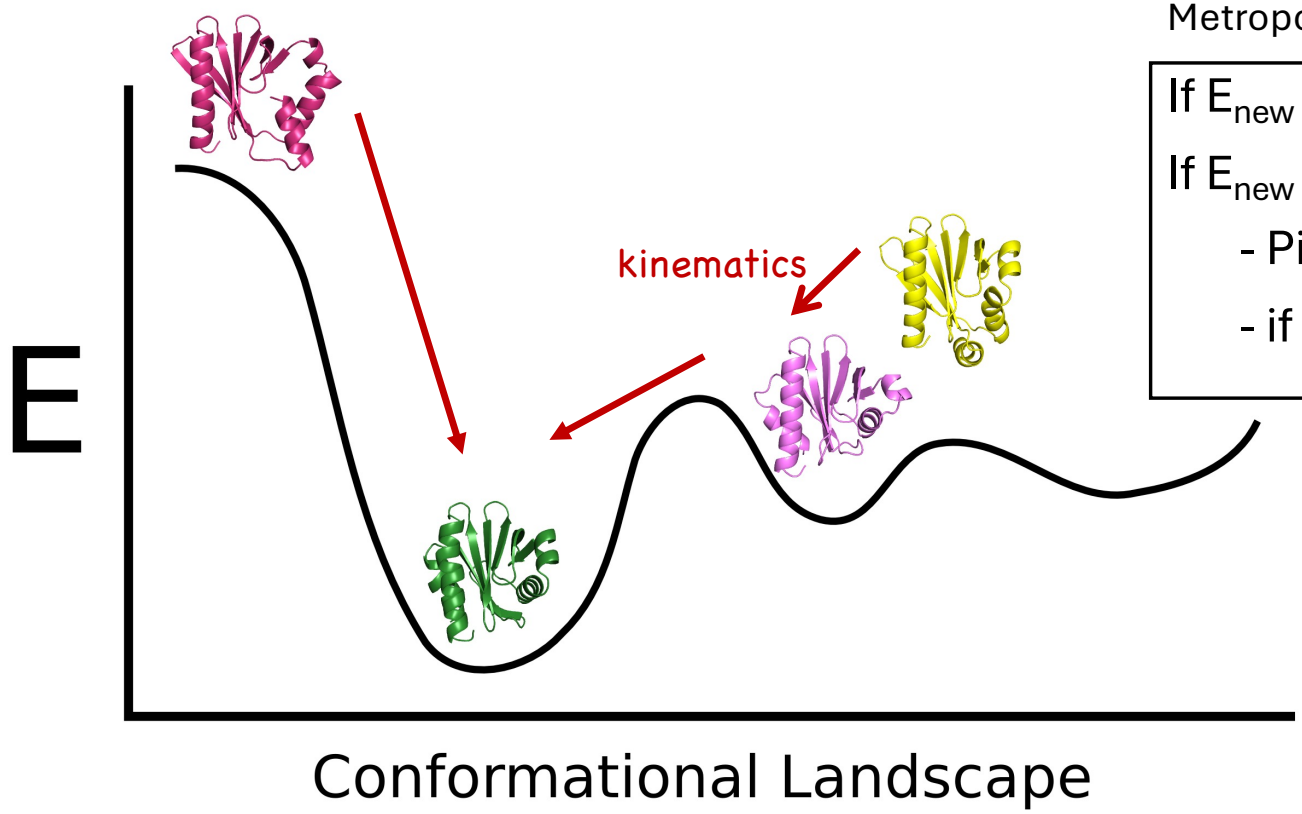
- Kinematics is the study of the motion of points, objects, and groups of objects without considering the causes of its motion.
- The primary degree of freedoms(DOFs) during Rosetta Monte Carlo simulation are dihedral angles, rather than the xyz coordinates of individual atoms (common in MD simulation).
- Thus, one must explicitly define the path by which internal coordinate changes propagate through the system. This path is referred to as the kinematic connectivity.
- Rosetta uses a tree structure to define the propagation of movement: AtomTree.

AtomTree in Rosetta

Tree: connected, acyclic graph
nodes → atoms
edges → kinematic connections



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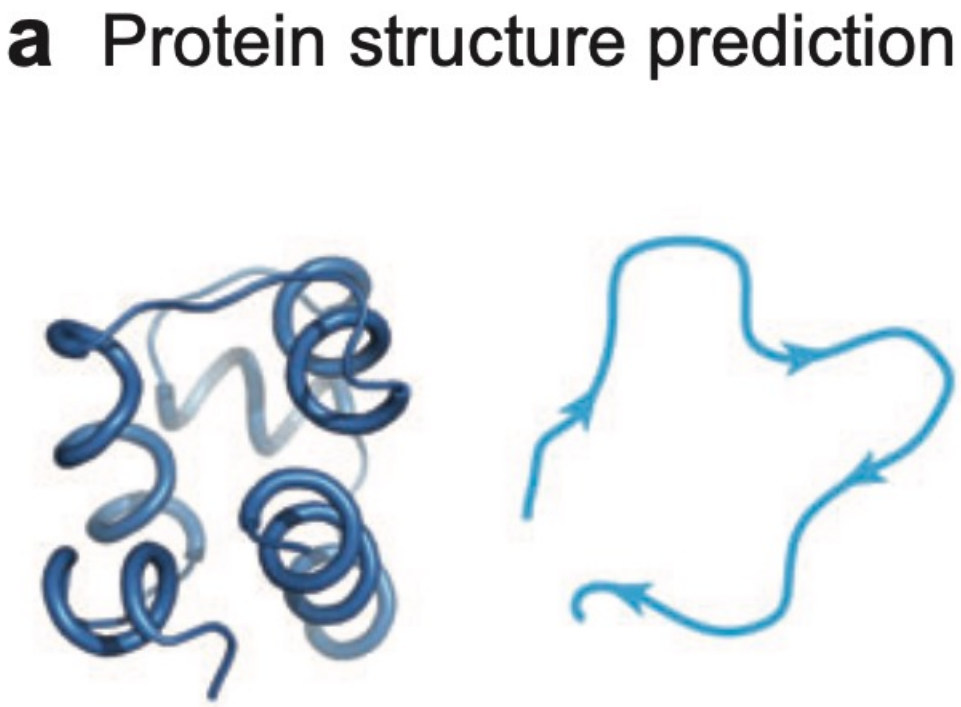
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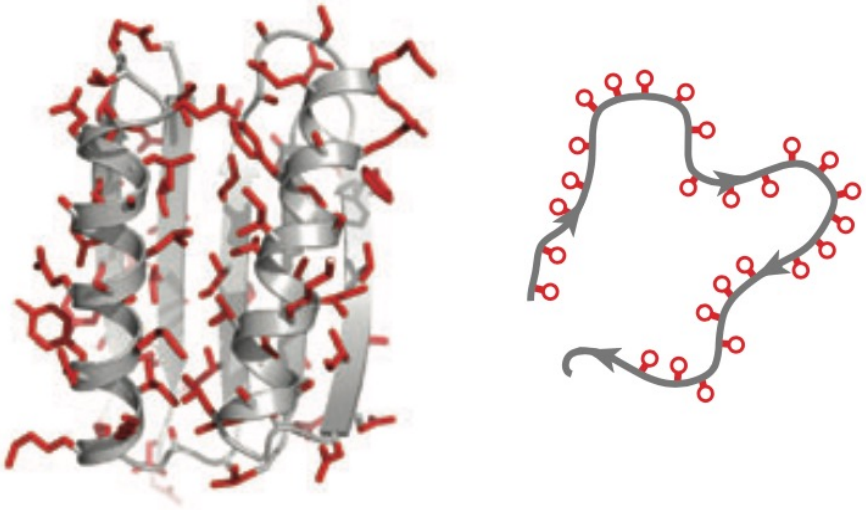
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Different kinematic connectivity

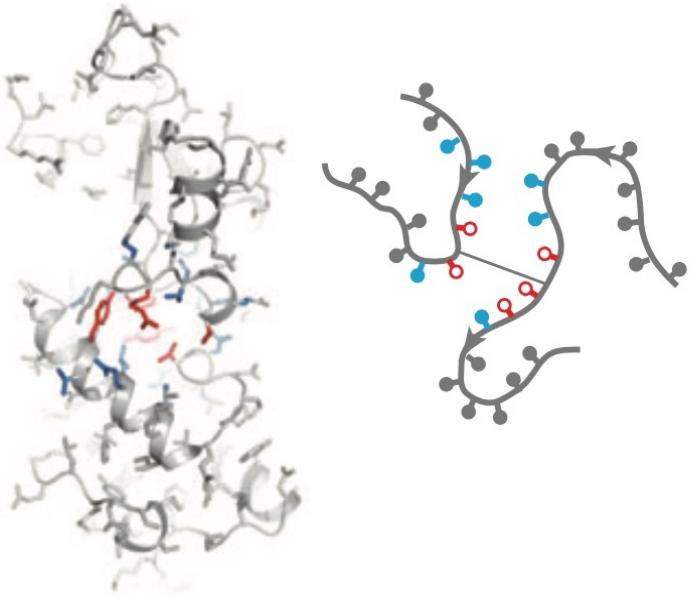


Different kinematic connectivity

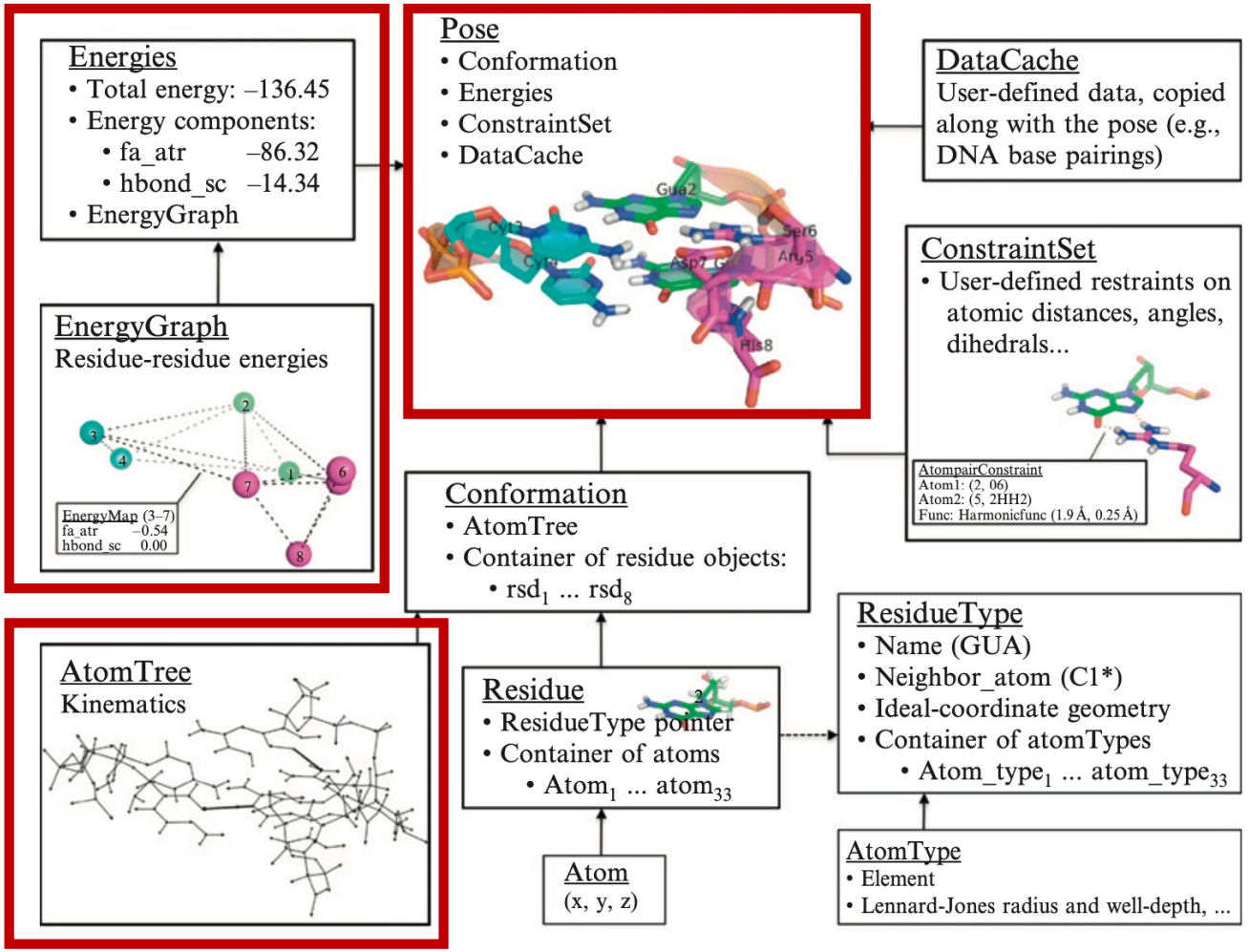
Protein design



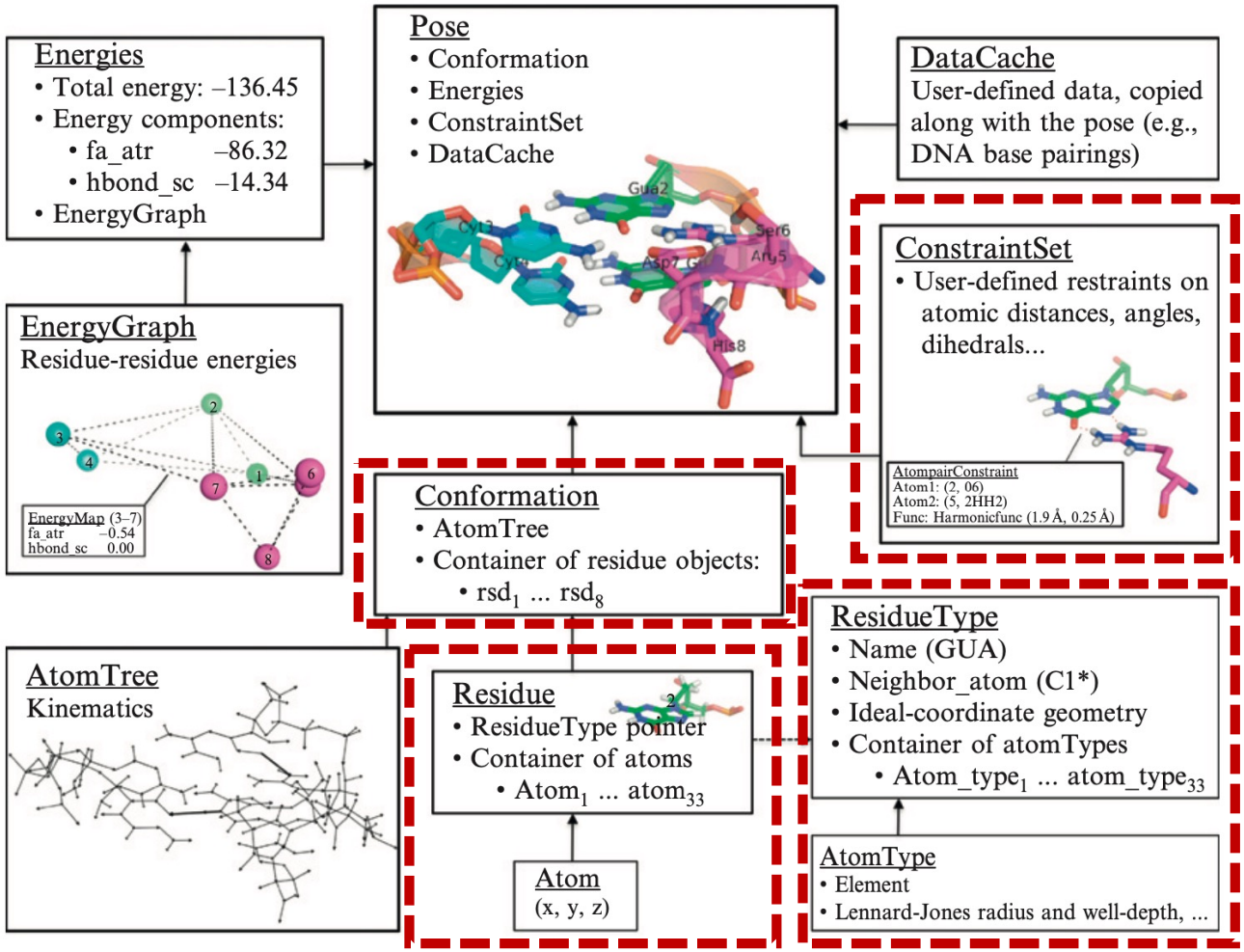
Protein-protein interface design



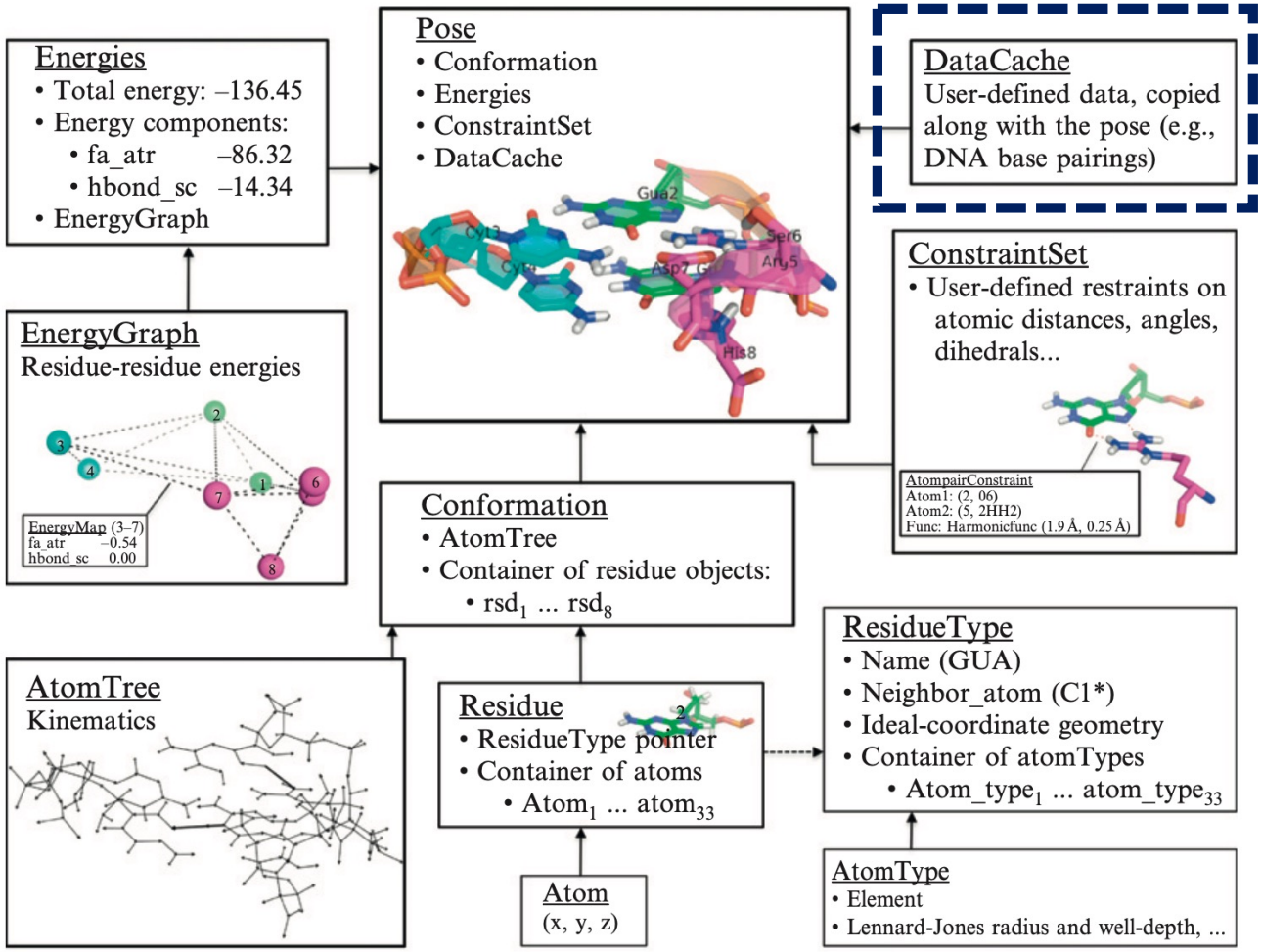
What is inside Rosetta?



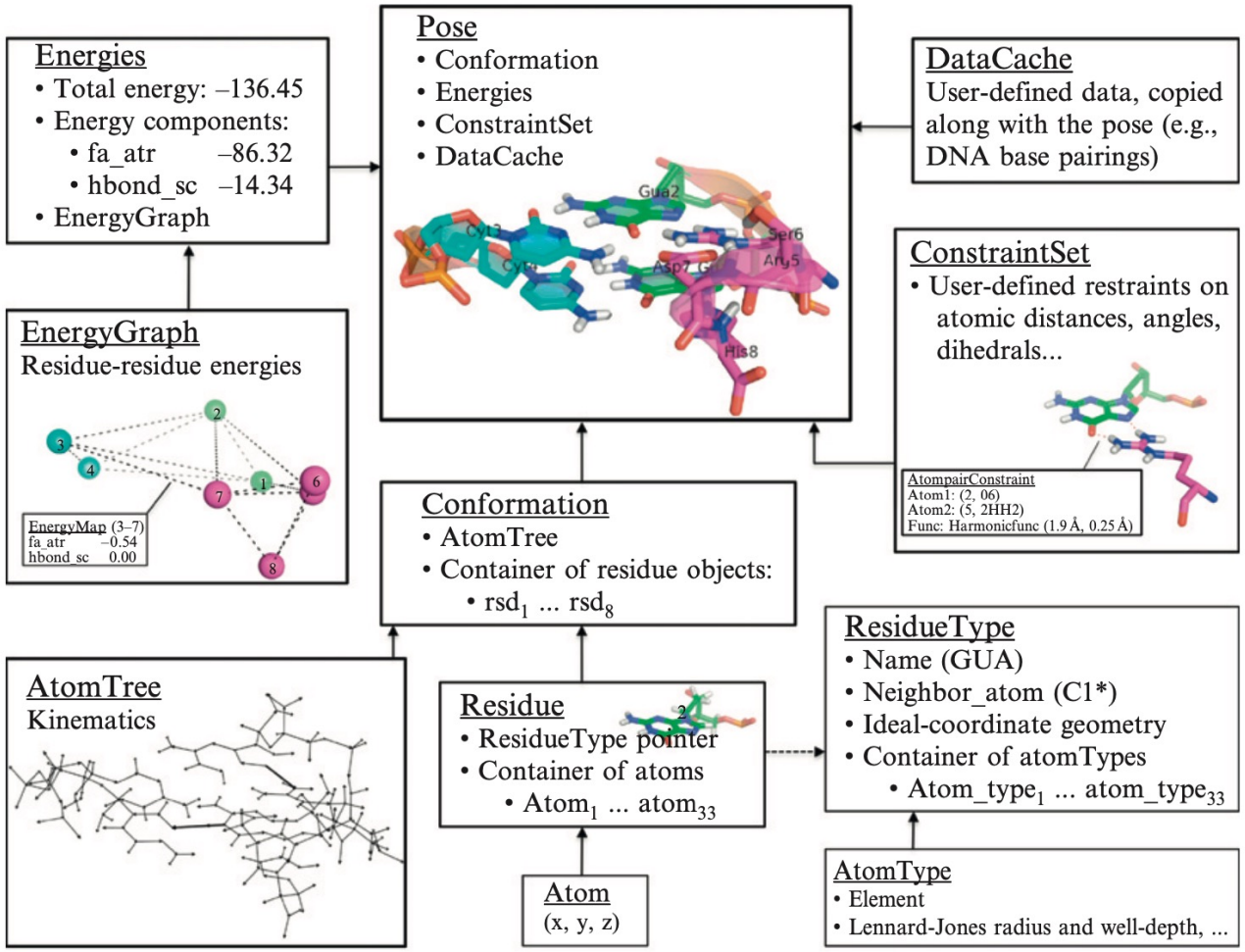
What is inside Rosetta?



What is inside Rosetta?



What is inside Rosetta?



Outline

- What is Rosetta?
- What can Rosetta do?
- What is inside Rosetta?
- How to use Rosetta?

How to Use Rosetta?

Foldit : Graphic User Interface(GUI)

PyRosetta: python scripting interface

Chaudhury, Sidhartha, Sergey Lyskov, and Jeffrey J. Gray. "PyRosetta: a script-based interface for implementing molecular modeling algorithms using Rosetta." *Bioinformatics* 26.5 (2010): 689-691.

Rosetta applications: C++ programs

Leaver-Fay, Andrew, et al. "ROSETTA3: an object-oriented software suite for the simulation and design of macromolecules." *Methods in enzymology* 487 (2011): 545.

RosettaScripts: xml scripting interface

Fleishman, Sarel J., et al. "RosettaScripts: a scripting language interface to the Rosetta macromolecular modeling suite." PloS one 6.6 (2011): e20161.

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